Quantum mechanics also has a Hamiltonian description that concentrates on incremental changes. It's called the time-dependent Schrödinger equation, and it's very general. As far as we know, it can be used to describe all physical systems. Still, it seems fair to ask, as Richard Feynman did almost seventy years ago, whether there is a way to look at quantum mechanics that pictures whole histories. In other words, is there a formulation that parallels the Principle of Least Action? I will not explain Feynman's path integral description in detail in this lecture, but just to whet your appetite I'll give you a hint of how it works.

First, let me very briefly remind you of the classical least action principle as I explained it in *Volume I*. Suppose that a classical particle starts at position x_1 at time t_1 and arrives at position x_2 at time t_2 (Fig. 9.5). The question is: What is the trajectory that it took between t_1 and t_2?

According to the least action principle, the actual trajectory is the one of minimum action. *Action* is of course a technical term, and it stands for the integral of the Lagrangian between the end points of the trajectory. For simple systems, the Lagrangian is the kinetic energy minus the potential energy. Thus, for a particle that moves in one dimension, the action is

$$A = \int_{t_1}^{t_2} L(x, \dot{x}) dt \qquad (9.19)$$

or

$$A = \int_{t_1}^{t_2} \left(\frac{m\dot{x}^2}{2} - V(x) \right) dt.$$

The idea is to try out all possible trajectories connecting the two end points, and calculate A for each one of them. The winner is the one that has the least action.[8,9]